Lecture 15

Metrics, the Hodge star and Maxwell's equations

Objectives. *A metric measures tangent vectors and selects a volume form once an orientation is chosen. The Hodge star then turns a form into a form of complementary degree. On spacetime this operation packages Maxwell's four equations into two equations for forms. The monopole flux shows why closed field forms need not have global potentials.*

By the end of this lecture you will be able to:

- Construct a Riemannian metric and lower an index using its bilinear form.
- Compute the Hodge star with the metric, orientation and signature specified.
- Derive all four Maxwell equations and charge conservation from the form equations.
- Use a nonzero sphere period to rule out a global monopole potential.

15.1 Metrics and the identification of vectors with covectors

Let M be a smooth n-manifold. A *Riemannian metric* g is a smoothly varying positive-definite symmetric bilinear form g_p on each tangent space T_pM. A *pseudo-Riemannian metric* requires nondegeneracy instead of positive definiteness. Nondegeneracy means that $g_p(v, w) = 0$ for all $w \in T_pM$ implies $v = 0$. We use g for the metric throughout Part II.

By Sylvester's law of inertia, a nondegenerate symmetric form admits a basis with diagonal entries $+1$ and -1. Such a basis is called *pseudo-orthonormal*. The two counts are independent of the basis. This is the signature result imported from *Linear Algebra for Physics*, Lecture 11, the example *Signature of a form*. Write its signature as $(n - s, s)$, with s negative directions. For a smooth nondegenerate metric, the signature is constant on each connected component: eigenvalues vary continuously and cannot cross zero. Here a *Lorentzian metric* in dimension $n \geq 2$ has signature $(1, n - 1)$.

In coordinates, with $i, j \in \{1, \dots, n\}$, $g = g_{ij}\,\mathrm{d}x^i \otimes \mathrm{d}x^j$. Its inverse matrix is written (g^{ij}). For $X \in T_pM$ and $\alpha \in T_p^*M$, nondegeneracy gives the mutually inverse *musical maps*

$$X^\flat = g(X, \cdot), \qquad (X^\flat)_i = g_{ij}X^j, \qquad (\alpha^\sharp)^i = g^{ij}\alpha_j.$$

These really lower and raise an index using a metric. Lecture 7's translation of sibling coordinate labels did not do this. For example, in one tangent space take the following metric matrix and vector components. The resulting lower components are

$$(g_{ij}) = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \qquad (X^i) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad ((X^\flat)_i) = \begin{pmatrix} 4 \\ 7 \end{pmatrix}.$$

The vector has squared length $4+14 = 18$. The figure shows how a positive metric determines its unit vectors.

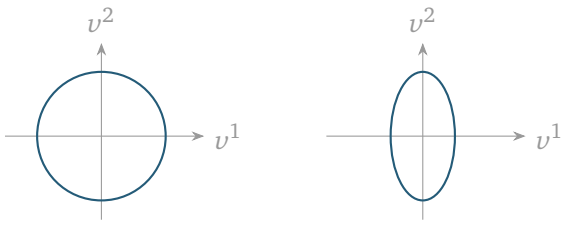


Figure 15.1: At a point, Euclidean unit vectors form a circle. For $g = 4\mathrm{d}x \otimes \mathrm{d}x + \mathrm{d}y \otimes \mathrm{d}y$ they form an ellipse with horizontal radius $1/2$.

Proposition 15.1 (Existence and induced metrics · *proved given partitions of unity*)**.** *Every smooth manifold admits a Riemannian metric. An immersion into a Riemannian manifold induces a Riemannian metric by pullback.*

Proof. On each coordinate domain use $g_a = \sum_i \mathrm{d}x_a^i \otimes \mathrm{d}x_a^i$. Choose a subordinate partition ρ_a from Theorem 12.1 and put $g = \sum_a \rho_a g_a$, extending each summand by zero. Supports make the extensions smooth. Local finiteness makes the sum smooth. For nonzero v, every summand $\rho_a g_a(v, v)$ is nonnegative, and at least one is positive because $\sum_a \rho_a = 1$. Thus the metric is positive definite. For an immersion f, the differential is injective, so $(f^*g)(v, v) = g(\mathrm{d}f(v), \mathrm{d}f(v)) > 0$ for $v \neq 0$. Smoothness and symmetry are preserved by pullback. ∎

An *isometry* is a diffeomorphism $f\colon (M, g_M) \to (N, g_N)$ with $f^*g_N = g_M$. A pseudo-Riemannian metric can restrict degenerately: for the matrix $\mathrm{diag}(1, -1, -1, -1)$, the vector $(1, 1, 0, 0)$ spans a line on which the restricted form is zero. So the induced-metric assertion used positivity.

15.2 Volume forms and the Hodge star

On an oriented pseudo-Riemannian n-manifold, the metric volume form is, in positive coordinates,

$$\mu = \sqrt{|\det(g_{ij})|}\, \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

The defining normalization is $\mu = \theta^1 \wedge \cdots \wedge \theta^n$ when $\theta^1, \dots, \theta^n$ is dual to a positive pseudo-orthonormal basis. To justify the coefficient, choose such a basis with metric matrix $S = \mathrm{diag}(\varepsilon_1, \dots, \varepsilon_n)$, where $\varepsilon_i = \pm 1$. If the coordinate basis has columns A in that basis, its metric matrix is $A^T S A$. Hence $|\det g| = (\det A)^2$. The dual bases satisfy $\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n = (\det A)^{-1}\theta^1 \wedge \cdots \wedge \theta^n$. Since $\det A > 0$, the coefficient gives the required normalization. Under a positive coordinate change the square root of the absolute metric determinant contributes the absolute Jacobian and the top wedge contributes its inverse. The expressions therefore agree. They are smooth because the determinant never vanishes. This is the chosen volume form μ of Lecture 11, now fixed by metric and orientation. Without orientation the same coefficient defines a density.

For $0 \le k \le n$, the inverse metric gives a bilinear pairing of covectors. Its induced pairing on decomposable k-forms is

$$\langle \alpha_1 \wedge \cdots \wedge \alpha_k, \beta_1 \wedge \cdots \wedge \beta_k \rangle_g = \det\big[g^{-1}(\alpha_i, \beta_j)\big]_{i,j=1}^{k},$$

extended bilinearly. Expanding each covector in a basis identifies this determinant with the bilinear extension of its values on basis wedges. On functions it is ordinary multiplication. This uses Lecture 10's determinant wedge, with no additional normalization from the fermionic wedge of *Quantum Mechanics I*. The pairing is nondegenerate: in a pseudo-orthonormal dual basis the basis wedges are orthogonal and each has norm squared +1 or −1.

Theorem 15.2 (Hodge star). *Let (M, g) be an oriented pseudo-Riemannian n-manifold and $0 \le k \le n$. There is a unique pointwise real-linear operator, smooth in position, $\star\colon \Omega^k(M) \to \Omega^{n-k}(M)$ such that*

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \mu \qquad (\alpha, \beta \in \Omega^k(M)).$$

If the metric has s negative directions, then on k-forms

$$\star\star\alpha = (-1)^{k(n-k)+s}\alpha.$$

Proof. The wedge pairing between degrees k and $n-k$ is nondegenerate: pairing with a basis wedge selects the coefficient of its complementary wedge. It therefore represents the functional $\alpha \mapsto \langle \alpha, \beta \rangle_g \mu$ uniquely. The coordinate linear system uses smooth inverse-metric coefficients and nonzero volume coefficient, so its solution varies smoothly.

For the sign, use a positively oriented pseudo-orthonormal dual basis $\theta^1, \dots, \theta^n$ at one point. For an increasing index set I, write θ^I for its wedge, I^c for its ordered complement, $\varepsilon_I = \prod_{i \in I} \varepsilon_i$, and σ_I for the sign of the permutation (I, I^c). The defining equation gives

$$\star\theta^I = \varepsilon_I \sigma_I \theta^{I^c}.$$

The two metric signs multiply to $\prod_i \varepsilon_i = (-1)^s$. Interchanging the blocks I and I^c uses $k(n-k)$ transpositions, so $\sigma_I \sigma_{I^c} = (-1)^{k(n-k)}$. Applying the formula twice proves the square identity on every basis wedge, hence on every form. ∎

For example, in oriented Euclidean $\mathbb{R}^3$, $\star dx = dy \wedge dz$ and $\star(dy \wedge dz) = dx$. The complementary plane is shown in the figure.

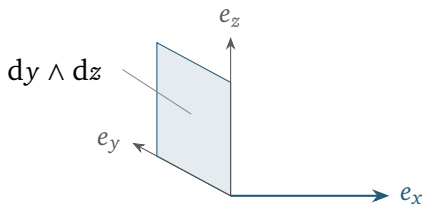


Figure 15.2: In Euclidean three-space, the covector dx pairs with the x direction. Its star is the oriented area form $dy \wedge dz$ on the complementary plane.

Reversing orientation negates μ and therefore negates the star. For a compactly supported real form α, the integral $\int_M \alpha \wedge \star\alpha$ is unchanged: the integrand and the integration orientation both change sign. The pairing need not be positive in indefinite signature. For complex forms extend the star complex-linearly and use $\bar{\alpha} \wedge \star\beta$ for the Hermitian pairing, antilinear in its first argument.

15.3 Maxwell's equations as equations for forms

Use coordinates $(x^0, x^1, x^2, x^3) = (t, x, y, z)$ on an open subset of spacetime $\mathbb{R}^4$. Greek indices run from 0 to 3 and spatial indices from 1 to 3. Take the Minkowski metric and orientation

$$\eta = \operatorname{diag}(1, -1, -1, -1), \qquad \mu = \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z.$$

This is the metric recalled from *Symmetries in Physics*, Lecture 17, *Minkowski space from physics*. Units have $c = 1$ and rationalized sources. The metric has $s = 3$, so $\star^2 = -1$ on 2-forms. A real 2-form satisfying $\star F = F$ or $\star F = -F$ must therefore be zero. Euclidean four-space instead has $\star^2 = +1$ on 2-forms.

Let the electric field $\mathbf{E}$, magnetic field $\mathbf{B}$, charge density ρ and spatial current $\mathbf{J}$ be smooth fields. Their Cartesian spatial components use the Euclidean spatial metric. They are not obtained by lowering spacetime indices with η. The real *field-strength 2-form* and the two *current forms* are

$$F_{\mathrm{em}} = E_i \mathrm{d}x^i \wedge \mathrm{d}t + B_x \mathrm{d}y \wedge \mathrm{d}z + B_y \mathrm{d}z \wedge \mathrm{d}x + B_z \mathrm{d}x \wedge \mathrm{d}y,$$
$$j = \rho \mathrm{d}t - J_i \mathrm{d}x^i, \qquad \mathcal{J} = \star j.$$

The subscripts em distinguish these real electromagnetic forms from later connection forms.

Theorem 15.3 (Maxwell's equations in forms). *On the specified spacetime, with the fields and units above, Maxwell's equations*

$$\operatorname{div}\mathbf{B} = 0, \quad \partial_t \mathbf{B} + \operatorname{curl}\mathbf{E} = 0, \qquad \operatorname{div}\mathbf{E} = \rho, \quad \operatorname{curl}\mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}$$

are equivalent to

$$\mathrm{d}F_{\mathrm{em}} = 0, \qquad \mathrm{d}\star F_{\mathrm{em}} = \mathcal{J}.$$

These equations imply charge conservation, $\partial_t \rho + \operatorname{div}\mathbf{J} = 0$. When these equations hold, every point has a neighbourhood with $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$ for a real 1-form A_{em}.

Proof. Use the star formula on the coordinate basis to obtain

$$\star F_{\mathrm{em}} = B_i \mathrm{d}t \wedge \mathrm{d}x^i + E_x \mathrm{d}y \wedge \mathrm{d}z + E_y \mathrm{d}z \wedge \mathrm{d}x + E_z \mathrm{d}x \wedge \mathrm{d}y,$$
$$\mathcal{J} = \rho \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z - J_x \mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z + J_y \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}z - J_z \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y.$$

For a spatial vector V, write the spatial 2-form $\mathcal{B}(V) = V_x \mathrm{d}y \wedge \mathrm{d}z + V_y \mathrm{d}z \wedge \mathrm{d}x + V_z \mathrm{d}x \wedge \mathrm{d}y$. Differentiation gives

$$\mathrm{d}F_{\mathrm{em}} = (\operatorname{div}\mathbf{B})\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z + \mathrm{d}t \wedge \mathcal{B}(\partial_t \mathbf{B} + \operatorname{curl}\mathbf{E}),$$
$$\mathrm{d}\star F_{\mathrm{em}} = (\operatorname{div}\mathbf{E})\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z + \mathrm{d}t \wedge \mathcal{B}(\partial_t \mathbf{E} - \operatorname{curl}\mathbf{B}).$$

For instance, the coefficient of $\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z$ in the first equation is $\partial_t B_x + \partial_y E_z - \partial_z E_y$. The other cyclic coefficients give the other components of the curl. Independence of the four basis 3-forms gives the four stated equations. Applying d to the source equation gives $\mathrm{d}\mathcal{J} = 0$. Differentiating the displayed current form gives $\mathrm{d}\mathcal{J} = (\partial_t \rho + \operatorname{div}\mathbf{J})\mu$, which proves conservation. Finally choose a star-shaped coordinate neighbourhood and apply Theorem 13.2 to the closed 2-form. Its primitive is the required local potential. ∎

Write the local potential in terms of its scalar and spatial parts as

$$A_{\mathrm{em}} = -\phi \mathrm{d}t + A_i \mathrm{d}x^i.$$

Taking its derivative gives $E_i = -\partial_i\phi - \partial_t A_i$ and $\mathbf{B} = \operatorname{curl}\mathbf{A}$, where $\mathbf{A} = (A_x, A_y, A_z)$. For a smooth real function χ, the *gauge transformation* $A_{\mathrm{em}} \mapsto A_{\mathrm{em}} + \mathrm{d}\chi$ leaves F_{em} unchanged because $\mathrm{d}^2\chi = 0$. Its component form is $A_i \mapsto A_i + \partial_i\chi$ and $\phi \mapsto \phi - \partial_t\chi$. This recalls *Theoretical Mechanics*, Lecture 3, *The charged particle*: the same transformation changes the particle Lagrangian by a total derivative.

Source signs must be translated together. Nakahara's Example 7.16 uses signature $-+++$.[1] The lower potential $\phi\mathrm{d}t - A_i\mathrm{d}x^i$ of Landau–Lifshitz has the opposite sign to A_{em} at matched units.[2]

15.4 The monopole obstruction

Let $r = \sqrt{x^2+y^2+z^2}$ and let $\Phi_{\mathrm{m}} \in \mathbb{R}\setminus\{0\}$ be the magnetic flux. A static *magnetic monopole* field on $\mathbb{R}\times(\mathbb{R}^3\setminus\{0\})$ has magnetic field $\mathbf{B} = \Phi_{\mathrm{m}}(x,y,z)/(4\pi r^3)$ and zero electric field. Its field 2-form is

$$F_{\mathrm{em}} = \frac{\Phi_{\mathrm{m}}}{4\pi}\,\frac{x\,\mathrm{d}y\wedge\mathrm{d}z + y\,\mathrm{d}z\wedge\mathrm{d}x + z\,\mathrm{d}x\wedge\mathrm{d}y}{r^3}.$$

Proposition 15.4 (No global monopole potential · *proved given partitions of unity*)**.** *This form is closed and has period Φ_{m} on the outward unit sphere in each fixed-time slice. It has local potentials but no global 1-form potential on the punctured spacetime.*

Proof. The form is time-independent, with no time differential. Its spatial part is the Coulomb flux form of § 12.7 times $\Phi_{\mathrm{m}}/(4\pi)$. Lecture 12 proved closedness and sphere period 4π for that form, giving the stated period. Local potentials follow from the Poincaré lemma. If $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$ globally, restriction to a fixed-time sphere and Stokes would give zero period, contradicting $\Phi_{\mathrm{m}} \neq 0$. ∎

No integer condition has been imposed on Φ_{m}. Lecture 22 studies the additional condition from a bundle and a charged particle.

15.5 Harmonic representatives

On a closed oriented Riemannian manifold, real forms of equal degree have the integrated pairing $(\alpha,\beta)_{L^2} = \int_M \alpha\wedge\star\beta$. The *codifferential* $\delta\colon \Omega^k(M)\to\Omega^{k-1}(M)$ is the formal adjoint of exterior differentiation, characterized by $(\mathrm{d}\alpha,\beta)_{L^2} = (\alpha,\delta\beta)_{L^2}$. This operator differs from the connecting map δ_{MV} of Lecture 13. The *Hodge Laplacian* is $\Delta = \mathrm{d}\delta + \delta\mathrm{d}$. A form is *harmonic* if $\Delta\alpha = 0$. Existence of this adjoint and its local formula are supplied by the star and integration by parts; see Nakahara, Definition 7.6 and Theorem 7.5. Sheet 15 derives that formula and computes the sphere Laplacian.

Theorem 15.5 (Hodge representatives · *cited*)**.** *Every de Rham class on a closed oriented Riemannian manifold has a unique harmonic representative.*

[1] For fixed orientation, metric reversal multiplies the k-form pairing and its star by $(-1)^k$. Thus the star on 2-forms is unchanged, while Nakahara's current is $j_{\mathrm{src}} = -j$. It follows that $\star_{\mathrm{src}} j_{\mathrm{src}} = \mathcal{J}$. His matrix (5.73), with its $(0,3)$ entry corrected to $-E_z$ by antisymmetry, represents F_{em}. His source equation is $\mathrm{d}^\dagger F_{\mathrm{em}} = j_{\mathrm{src}}$. Definition 7.6 gives $\mathrm{d}^\dagger = \star_{\mathrm{src}}\mathrm{d}\star_{\mathrm{src}}$ on Lorentzian 2-forms. Apply the source star and use its square +1 on 3-forms to obtain $\mathrm{d}\star F_{\mathrm{em}} = \mathcal{J}$.

[2] At matched units, $A^{\mathrm{LL}} = -A_{\mathrm{em}}$ and $F^{\mathrm{LL}} = -F_{\mathrm{em}}$. To convert instead to Gaussian fields with $c=1$, set $F_{\mathrm{G}} = \sqrt{4\pi}F_{\mathrm{em}}$ and $\mathcal{J}_{\mathrm{G}} = \mathcal{J}/\sqrt{4\pi}$. Then $\mathrm{d}\star F_{\mathrm{G}} = 4\pi\mathcal{J}_{\mathrm{G}}$. The Gaussian lower-potential convention $F_{\mathrm{G}}^{\mathrm{LL}} = -F_{\mathrm{G}}$ therefore gives $\mathrm{d}\star F_{\mathrm{G}}^{\mathrm{LL}} = -4\pi\mathcal{J}_{\mathrm{G}}$. This converts the field sign and source normalization together.

The analytic existence theorem is cited from Nakahara, Theorem 7.8, §7.9.6. The closedness hypothesis is inherited there from Theorem 7.7. The statement does not apply to the Lorentzian spacetime used above.

15.6 The spacetime star and vacuum duality

The six basis 2-forms give a direct way to use the signature convention. For example, $dt \wedge dx$ has norm squared -1 and $(dt \wedge dx) \wedge (dy \wedge dz) = \mu$. Its star is therefore $-dy \wedge dz$. The same rule gives the table:

α	$\star\alpha$	α	$\star\alpha$
$dt \wedge dx$	$-dy \wedge dz$	$dy \wedge dz$	$dt \wedge dx$
$dt \wedge dy$	$+dx \wedge dz$	$dx \wedge dz$	$-dt \wedge dy$
$dt \wedge dz$	$-dx \wedge dy$	$dx \wedge dy$	$dt \wedge dz$

Example 15.6 (An electric field and its magnetic dual)**.** Take a constant electric field $\mathbf{E} = E_0 e_x$, with $E_0 \neq 0$, and zero magnetic field on Minkowski space. Its potential and field are

$$A_{\mathrm{em}} = E_0 x\, dt, \qquad F_{\mathrm{em}} = E_0 dx \wedge dt.$$

All field derivatives vanish, so Maxwell's equations hold with zero sources. The table gives $\star F_{\mathrm{em}} = E_0 dy \wedge dz$. This is the field of a constant magnetic field $\mathbf{B}' = E_0 e_x$ and zero electric field. A potential for it is $E_0 y\, dz$. Thus the transformation $F_{\mathrm{em}} \mapsto \star F_{\mathrm{em}}$ changes this electric solution into a magnetic solution.

For any source-free field, this transformation is *vacuum duality*. Indeed, $dF_{\mathrm{em}} = 0$ and $d\star F_{\mathrm{em}} = 0$ imply $d(\star F_{\mathrm{em}}) = 0$ and $d\star(\star F_{\mathrm{em}}) = -dF_{\mathrm{em}} = 0$. The transformed components are $\mathbf{E}' = -\mathbf{B}$ and $\mathbf{B}' = \mathbf{E}$. Applying duality twice gives $-F_{\mathrm{em}}$, rather than F_{em}.

Self-check. In oriented Euclidean $\mathbb{R}^3 \setminus \{0\}$, take $\mathbf{E}^\flat = (x dx + y dy + z dz)/r^3$. Compute $\star\mathbf{E}^\flat$.

Answer. The Euclidean rules give

$$\star\mathbf{E}^\flat = \frac{x\, dy \wedge dz + y\, dz \wedge dx + z\, dx \wedge dy}{r^3}.$$

This is the Coulomb flux form called β in Lecture 12. Here the spatial Euclidean star is used, not the four-dimensional spacetime star.

Summary. A Riemannian metric exists by partitions of unity and identifies tangent vectors with covectors. A metric and orientation define volume and the Hodge star. Its square depends on degree, dimension and the number of negative metric directions. With signature $+---$, Maxwell's equations are $dF_{\mathrm{em}} = 0$ and $d\star F_{\mathrm{em}} = \mathcal{J}$. They imply charge conservation and local potentials. Monopole flux prevents a global potential. On closed oriented Riemannian manifolds, the cited Hodge theorem selects one harmonic form in each cohomology class. Lecture 16 uses a metric to differentiate tangent fields and define geodesics.

Reading. Nakahara, *Geometry, Topology and Physics*, second edition, §7.1 and §7.9, especially (7.172)–(7.177), Example 7.16, and Theorems 7.7–7.8. Tu, *Differential Geometry*, Theorem 1.12, pp. 6–7, and §16.4–§16.5, pp. 132–135.